Robustness of Implicit Q-Learning Under Controlled Distribution Shift

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Core Challenge

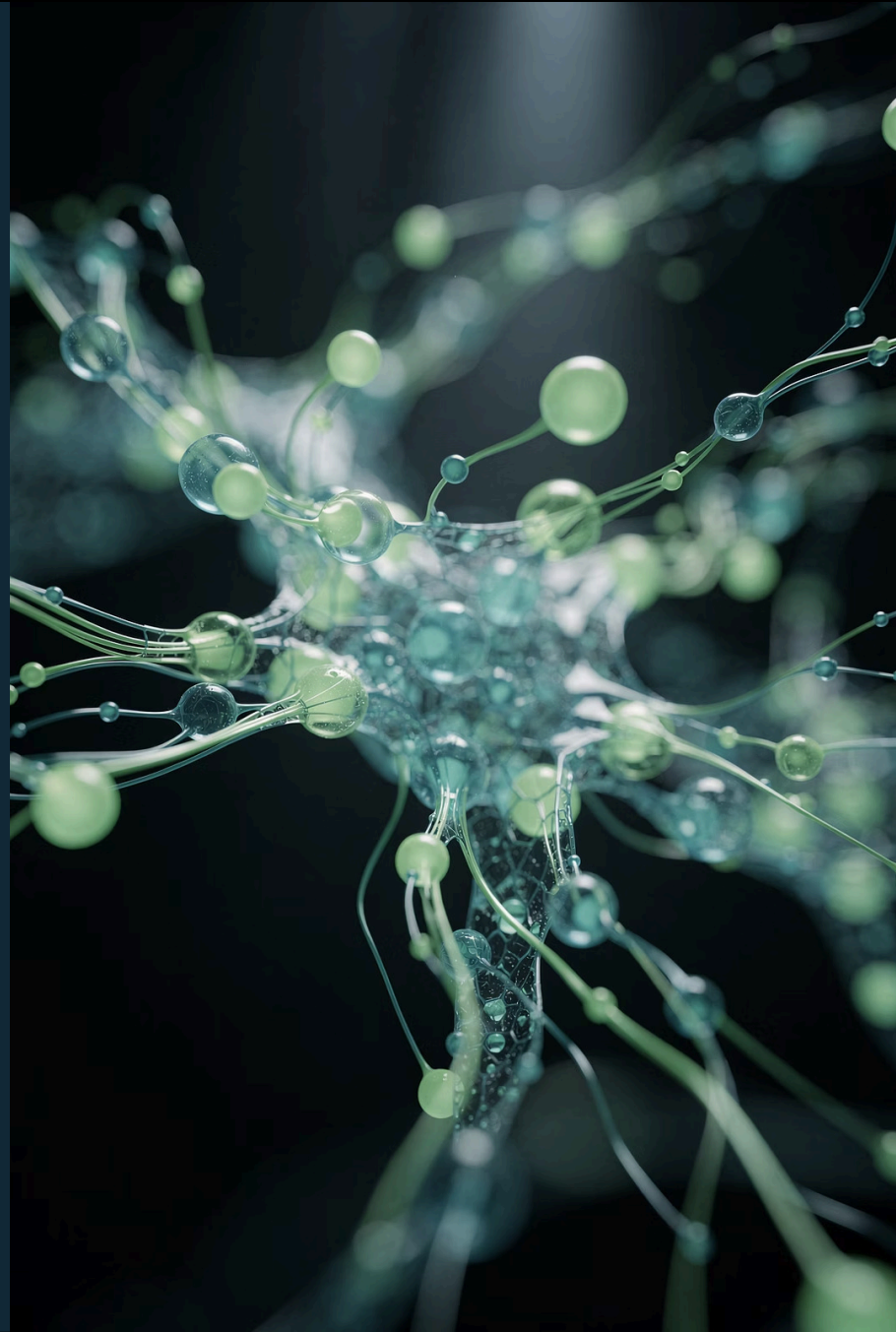
Offline RL must generalize under shift.

Algorithm Studied

IQL uses expectile and advantage-weighted updates.

Key Extension

TripleCritic and τ ablation study.



Background: Offline RL & Distribution Shift

Why Offline RL?

It is useful in safety-critical settings where interaction is costly or impossible. Models train on fixed datasets but must still generalize.

Research Question

How robust is IQL under controlled distribution shift?

Three Competing Strategies

→ **CQL**

Uses pessimism to reduce out-of-distribution Q-values.

→ **IQL**

Stays within the dataset support for updates.

→ **TD3+BC**

Uses behavior cloning to keep the policy close.

Methodology: IQL Reproduction & Shift Wrappers

We train IQL once on standard D4RL medium datasets for 300k steps, then apply wrappers only at test time.

Gravity Wrapper

Scales gravity from 0.5× to 2.0×.

Observation Noise

Adds Gaussian noise at four levels.

Friction Wrapper

Scales friction from 0.5× to 2.0×.

Reward Perturbation

Adds reward noise as the control.

Experimental Results: 2Q vs. 3Q Robustness

Baseline Scores

Environment	2Q (Baseline)	3Q (Ensemble)
Hopper-medium-v2	1712	1426
HalfCheetah-medium-v2	5510	5536
Walker2d-medium-v2	3550	3549

Hopper AUDC

Shift Type	2Q	3Q	Winner
Gravity	0.596	0.574	3Q
Obs Noise	0.146	0.142	3Q
Friction	0.738	0.687	3Q
Reward Perturb	0.002	0.001	3Q

Ablation Study: Expectile τ & Gravity Case Study

Hopper Gravity Shift

Gravity	2Q Score	3Q Score	2Q Drop%	3Q Drop%
0.5×	9.45	12.18	44.81	32.58
1.0×	57.66	43.84	0.00	0.00
1.5×	22.85	24.23	40.06	21.80
2.0×	10.92	11.81	50.77	33.37

Hopper AUDC by Expectile τ

τ	Config	Baseline	Gravity	Friction
0.5	2Q	1483	0.546	0.700
0.7	2Q	1712	0.596	0.738
0.8	2Q	2045	0.757	0.782
0.9	2Q	1603	0.668	0.714

Key Findings & Future Directions

Q-Ensemble is Environment-Dependent

3Q helps Hopper, but Walker2d and HalfCheetah vary.

Gravity & Friction Hurt Most

Reward perturbations have negligible impact.

Lower τ Improves Robustness

Choose τ based on deployment conditions.

Future Direction 1: Shift-Aware τ Selection

Adapt τ to the expected environment.

Future Direction 2: Adaptive Ensemble Sizing

Select critic count per environment and dataset.